Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/28/2024
PI Name:	McClure, Ryan Ph.D.		
Project Title:	Dynamics of Microbiomes in Space (DynaMoS)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ryan.Mcclure@pnnl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	Battelle Memorial Institute (Pacific Northwest National Laboratory)		
PI Address 1:	Biological Sciences Division, Earth and Biological Sciences		
PI Address 2:	902 Battelle Blvd, PO Box 999, MSIN J4-18		
PI Web Page:			
City:	Richland	**State:**	WA
Zip Code:	99354-1793	**Congressional District:**	4
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	02/07/2020	**End Date:**	12/31/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
Contact Email:	Denise.E.Freeland@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: This project is a continuation of the investigation titled, "Dynamics of Microbiomes in Space (DynaMoS)", with Dr. Ryan McClure from the Battelle Memorial Institute (Pacific Northwest National Laboratory) as the new Principal Investigator (PI). The previous PI was Dr. Janet Jansson at PNNL from February to June 2023.		
COI Name (Institution):	Hixson, Kim Ph.D. (Battelle Memorial Institute) Rivas-Ubach, Albert Ph.D. (Battelle Memorial Institute) Farris, Yuliya Ph.D. (Battelle Memorial Institute)		
Grant/Contract No.:	Department of Energy IAA		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to examine the population dynamics and community interactions of naturally co-adapted soil microbial consortia using multi-omics analysis, correlative molecular networking and metagenomics-based metabolic modeling, and compare results between the International Space Station (ISS) and ground control at Kennedy Space Center (KSC). We hypothesize that the selection pressure (altered atmospheric gas composition, microgravity, and increased radiation) imposed by the space station environment will alter both the microbial community population dynamics and the metabolic interactions between specific microbial community members.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Soil microorganisms are essential for life on our planet. They carry out key functions, including cycling carbon and other nutrients, and supporting plant growth. On Earth, soil microorganisms exist in communities that coordinate their metabolism to carry out different steps in complex metabolic processes. Our research is focused on a defined consortium of soil microorganisms that carry out steps required for decomposition of chitin--the second most abundant carbon polymer on Earth. It is not known how interspecies interactions may be impacted by the space environment. Therefore, our research will provide beneficial information about how soil microorganisms function in space and if their metabolism is altered when compared to normal conditions on Earth. Knowledge gained will be beneficial for future space missions that aim to achieve life-sustainable conditions that rely on natural processes carried out by soil microorganisms.
Task Progress:	New project for FY2023. The PI changed from Dr. Janet Jansson to Dr. Ryan McClure after Dr. Jansson retired from the Battelle Memorial Institute (Pacific Northwest National Laboratory/PNNL). Dr. McClure is based at PNNL. See the project under Dr. Jansson for prior reporting. The project under Dr. Jansson was also titled, "Dynamics of Microbiomes in Space (DynaMoS)". Over the last year, we have completed analysis of our Experiment Verification Test (EVT) data and have submitted the results as a publication to Communications Earth & Environment, a Nature journal. A summary of our EVT analysis is below; briefly, the data clearly and consistently revealed a temporal shift during chitin decomposition and defined contributions by individual species. A Streptomyces species was a key player in early steps of chitin decomposition, followed by other members of model soil consortium-2 (MSC-2). The species composition of MSC-2 also changed during the incubations. The MSC-2 species composition shifted initially, when comparing 0 to 4 weeks of incubation. This shift also included a 5-day preincubation at 4 °C to 20 °C, when the cell activity increased. At subsequent timepoints, the MSC-2 species composition continued to shift as the chitin was metabolized, but the magnitude of the change was less. To define which MSC-2 members were interacting with each other, (either negatively or positively), the individual species abundances were compared across timepoints and replicates. This analysis showed that Ensifer was strongly positively correlated with both Variovorax and Sphingopyxis, with the latter two species also strongly correlated with each other. In addition, Streptomyces and Neorhizobium were strongly correlated with each other. By contrast, Ensifer, Variovorax and Sphingopyxis were all negatively correlated with Streptomyces and Neorhizobium. Finally, Rhodococcus was negatively correlated with Ensifer, and Sinorhizobium was negatively correlated with Dyadobacter. All correlations described above were significant with a p-value < 0.05. These data revealed that the 8 species could be clustered into two groups that had similar correlation profiles; each group with four species. We saw similar patterns with our metatranscriptomic, metaproteomic and metabolomic analysis of MSC-2 across the four timepoints: major shift from 0 to 4 weeks, followed by more moderate shifts as the experiment proceeded to 12 weeks. We also saw differences in the identity of the most active MSC-2 member under different -omics analysis. Examining DNA, the community was fairly even, but examining RNA Rhodococcus was the most transcriptionally active; and via protein analysis, we found that Sphingopyxis was the most translationally active. Our multi-omic analysis also allowed us to determine a breakdown pathway for chitin, the primary carbon and nitrogen source in these experiments. We found that early stages of chitin breakdown were precipitated by Streptomyces, while middle stages of breakdown were the result of a combinatorial input from many species. Later stages of breakdown were driven by Ensifer and Sphingopyxis. The data above was from our EVT experiment, but we have also made great progress in analysis data from our full experiment comparing ground samples to those incubated aboard the International Space Station (ISS). When focusing on our 12 week samples, we find a clear delineation in samples incubated at NASA Kennedy Space Center (KSC) on the ground, compared to those incubated in the ISS when looking at DNA abundances of the four species. Importantly, we do not see this difference in samples at the start of the experiment, so the changes we are seeing are the result of time on the ISS, not an artifact of the experimental setup. We also see major differences in the rate of community change in ISS versus ground samples. Under ground conditions the rate of change for the communities across time is rapid; after four weeks, they move to their final state and do not change much in the subsequent weeks, similar to our EVT experiment which was also under ground conditions. However, aboard the ISS the rate of change is much slower; communities eventually get to their final state, but this happens gradually. We also saw changes in abundance for specific species, when comparing ISS to ground incubation. Ensifer showed a statistically significant increase in expression on board the ISS, while Rhodococcus showed a statistically significant decrease in expression. Changes between ISS and ground were also found in our metatranscriptomic analysis of samples. As with DNA, there were minimal changes in the RNA profiles at the start of the experiment (0 weeks) between ground and ISS. However, after 12 weeks, the transcriptomic profiles have shifted significantly and there is a clear difference between ground and ISS. We also looked at the transcriptomic profiles of individual species and found that Rhodococcus by far showed the greatest number of differentially expressed genes, when comparing ground to ISS. We note above that Rhodococcus also showed a drop in abundance on the ISS and the transcriptomic shift for this species may be related to that abundance drop. Over the last three years, we have had major success on the DynaMoS project. We have shown clearly that chitin degradation by a known community in soil is a process driven by division of labor, with certain species stepping in to do certain roles. Much of the early chitin degradation is driven by a single species, with a broader range of members stepping in for downstream breakdown. These initial experiments provide us with a strong foundation to better analyze and interpret our data collected from incubation on the ISS compared to ground controls, a process that is in the final stages now. We can already see major changes in the make-up of the community as a function of spaceflight, as well as the rate of community change; much slower on the ISS, and the expressed functions of the community at the RNA level, largely driven by changes in gene expression in Rhodococcus. All of this data supports one of our main hypotheses of this study: that incubations in space will have major effects on microbial community growth and function. As we continue to analyze this data we will be able to identify specific processes that show differences

	between space and ground controls, initially focusing on chitin breakdown, an approach made easier by our EVT data. The outcomes of that analysis will lead to new insights into how we can apply and harness microbial communities in space settings for carbon cycling, plant growth promotion and other needs of astronauts during spaceflight.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	McClure R, Rivas-Ubach A, Hixson KK, Farris Y, Garcia M, Danczak R, Davison M, Paurus VL, Jansson JK. "Multi-omics of a model bacterial consortium deciphers details of chitin decomposition in soil." mBio. 2025 Jul 9;16(7):e0040425. https://doi.org/10.1128/mbio.00404-25 ; PubMed PMID: 40444942; PubMed Central PMCID: PMC12239585 , Jul-2025